

Name: Sualden S. Sala

Course: ITD103 – Modern Query Languages

Lab 1, Part A: CAP Theorem Analysis

1. Online banking transaction system

- **CAP Priority:** Consistency (C) and Partition Tolerance (P).
- **Database Selection:** MongoDB (configured for strong consistency) or Neo4j.
- **Justification:** Financial systems cannot tolerate inaccurate balances. If a network partition occurs, the system must reject transactions rather than risk allowing users to act on outdated data (Consistency over Availability).

2. Social media feed

- **CAP Priority:** Availability (A) and Partition Tolerance (P).
- **Database Selection:** Cassandra.
- **Justification:** Users expect social media apps to load instantly and always accept new posts. It is perfectly acceptable if a new post takes a few extra seconds to sync globally to all users' feeds (Eventual Consistency), as long as the system remains highly available.

3. IoT sensor data collection

- **CAP Priority:** Availability (A) and Partition Tolerance (P).
- **Database Selection:** Cassandra or Redis.
- **Justification:** Sensors stream massive amounts of data continuously. The database must always be available to accept this incoming firehose of data. If a node goes down, the system must keep accepting data to avoid losing readings.

4. E-commerce shopping cart

- **CAP Priority:** Availability (A) and Partition Tolerance (P).
- **Database Selection:** Redis or Cassandra.

- **Justification:** Revenue is lost if a customer cannot add items to their cart. The system must remain highly available. If nodes are partitioned, it is better to temporarily have a slightly out-of-sync cart (resolved during the checkout phase) rather than preventing the user from shopping.